

Lab Experiment: 07

Assignment 1st: Linear Search Implementation

```
#include <stdio.h>
```

```
int linearSearch(int arr[], int size, int target) {
```

```
    for (int i = 0; i < size; i++) {
```

```
        if (arr[i] == target)
```

```
            return i;
```

```
    }
```

```
    return -1; }
```

```
int main() {
```

```
    int size, target, result, arr[size];
```

```
    printf("Enter the number of elements in the array: ");
```

```
    scanf("%d", &size);
```

```
    printf("Enter the elements of the array:\n");
```

```
    for (int i = 0; i < size; i++)
```

```
        scanf("%d", &arr[i]);
```

```
    printf("Enter the element to search for: ");
```

```
    scanf("%d", &target);
```

```
    result = linearSearch(arr, size, target);
```

```
    if (result != -1)
```

```
        printf("Element found at index %d.\n", result);
```

```
    else
```

```
        printf("Element not found in the array.\n");
```

```
    return 0; }
```

```
Enter the number of elements in the array: 4
Enter the elements of the array:
31
61
91
11
Enter the element to search for: 91
Element found at index 2.

=== Code Execution Successful ===
```

Assignment 2nd: Binary Search Implementation

```
#include <stdio.h>
```

```
// Function to perform binary search
```

```
int binarySearch(int arr[], int size, int target) {
```

```
    int left = 0, right = size - 1;
```

```
    while (left <= right) {
```

```
        int mid = left + (right - left) / 2;
```

```
        // Check if target is present at mid
```

```
        if (arr[mid] == target)
```

```
            return mid;
```

```
        // If target is greater, ignore the left half
```

```
        if (arr[mid] < target) {
```

```
            left = mid + 1;
```

```
        }
```

```
        // If target is smaller, ignore the right half
```

```
        else {
```

```
            right = mid - 1;
```

```

    }
}
return -1; // Return -1 if the target is not found
}

int main() {
    int size, target, result, arr[size];

    printf("Enter the number of elements in the array: ");
    scanf("%d", &size);

    // Input: Array elements (must be sorted)
    printf("Enter the elements of the sorted array:\n");
    for (int i = 0; i < size; i++) {
        scanf("%d", &arr[i]);
    }

    // Input: Target element to search for
    printf("Enter the element to search for: ");
    scanf("%d", &target);

    result = binarySearch(arr, size, target);
    if (result != -1) {
        printf("Element found at index %d.\n", result);
    } else {
        printf("Element not found in the array.\n");
    }

    return 0;
}

```

```
Enter the number of elements in the array: 4
Enter the elements of the sorted array:
22
30
33
40
Enter the element to search for: 33
Element found at index 2.
```

```
=== Code Execution Successful ===
```